An audited reimplementation of RMT-based molecular ecological network analysis

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Code and audit trail: github.com/RaphaelRibes/hydro-mena (commit `8ae7c7c`)

Abstract — We reimplemented the Molecular Ecological Network Analysis (MENA) pipeline of Deng et al. (2012) in Rust and audited the inherited 2025 code before trusting its outputs. The audit found that the random-matrix-theory (RMT) threshold scan had never actually executed its goodness-of-fit test — every candidate was discarded by preprocessing, so an earlier "no threshold is estimable" conclusion was a bug report, not a finding. After repairing the unfolding, the eligibility floors and the modularity formula, the automatic threshold is estimable on two of three real hydrocarbon-degradation conditions (box: 0.680, ox: 0.610, anox: 0.650 on the binary adjacency spectrum — all three estimable; the weighted-spectrum mode, where ox reads Wigner-like at $\beta = 0.92$, is kept as an ablation). We add Louvain modularity detection, degree-preserving null z-scores, bootstrap module stability, signed edges, a Clauset scale-free test, and a centred log-ratio (CLR) option that makes the oxic condition estimable (st = 0.63). All networks show modularity 20–37 null standard deviations above degree-matched random graphs. The implementation is open source, 40/40 tests, with an exact CPU/GPU parity and a GPU path 2.8× faster than CPU at n = 12 000.

1 Introduction

Microbial communities are too diverse and too undersampled for pairwise association tests to be trustworthy on their own: with hundreds of OTUs and a few dozen samples, thousands of spurious correlations are inevitable. Deng et al. (2012) proposed to escape this by **not** choosing a correlation cutoff at all: build the weighted similarity matrix, threshold it at candidate values, and let random matrix theory decide which threshold produces a network whose adjacency spectrum is statistically indistinguishable from a random (Poisson) one. The pipeline — standardise, correlate, threshold, RMT-select, measure topology, detect modules, correlate module eigengenes with environmental traits — is the MENA pipeline, and the 2025 internship code in our lab was an implementation of it.

Before reusing inherited results we audited the implementation. The audit mattered: the inherited scan's χ^2 goodness-of-fit never ran, so its "not estimable" verdicts said nothing about the data. This paper reports (i) what was broken and how it was found, (ii) the corrected automatic thresholds on three real conditions, and (iii) the additions that make the method more robust — null models, bootstrap stability, signed edges, a defensible scale-free test, compositionality handling, and a GPU path. All numbers are reproducible from the repository; every run writes a per-candidate audit trail.

2 Methods

2.1 Data

Three conditions from a long-term hydrocarbon-degradation experiment ("RawStorage 2025" revision): **box** (178 OTUs × 18 samples), **oxic / ox** (173 × 36), **anoxic / anox** (181 × 34), each with 74 hydrocarbon traits (n-alkanes nC10–nC…, PAHs). OTU abundances are relative abundances per sample; taxonomy strings are carried through as node names.

2.2 Pipeline

1. **Standardise** each OTU row (z-score across samples). Optionally apply the centred log-ratio transform first (`--clr`, pseudocount 1.0):

$$\mathrm{clr}(x)_{ik} = \log(x_{ik} + pc) - \overline{\log(x_{i\cdot} + pc)}$$

. Relative abundances are compositional — each sample sums to a constant — so raw Pearson correlations are biased by the closure constraint; CLR removes it. Off by default; used for the ox sensitivity analysis.

2. **Similarity**: Pearson r between OTU rows; $S = |r|$ (Eq. 1–2 of Deng et al.), or signed r with `--signed` (sign kept as edge metadata; the RMT stage still thresholds $|r|$ because negative weights would break the weighted-adjacency Eq. 3).
3. **Threshold**: automatic RMT scan (default) or user-set `--st`. The scan walks st from 0.30 upward; at each candidate it builds the thresholded adjacency and eigen-decomposes its **binary** 0/1 form (default `--binary-spectrum`, matching the 2025 repo's `fig_5_threshold.py`: weighted correlation-thresholded spectra carry level repulsion — Brody β up to 0.92 on real data — so their NNSD is Wigner-like and never passes the Poisson GOF; binary random-graph spectra are Poisson. `--no-binary-spectrum` reverts to the weighted matrix of Deng Eq. 3 for comparison). The graph itself keeps the weights $a_{ij} = s_{ij}$ if $s_{ij} \geq s_t$ else 0 (Eq. 3); takes the spectrum of the largest connected component, unfolds by local density average (paper step f), computes nearest-neighbour spacings, and runs a binned χ^2 GOF against the Poisson law $\exp(-d)$ at α = 0.01 (paper step h). Eligibility floors, calibrated on simulations: ≥ 50 % of OTUs retained and ≥ 50 % of kept nodes in the giant component. First accepted candidate wins; if none is accepted the tool reports **not estimable** and writes no graph rather than inventing one.
4. **Topology**: nodes, edges, density, mean degree, geodesic distance, average clustering, modularity, connected-component stats. Modularity uses the full formula $Q = \sum_c \left(\frac{L_c}{2} E - \left(\frac{d_c}{2} E \right)^2 \right)$ — the configuration term is what makes a one-community complete graph score exactly 0.
5. **Null model** (`--null-samples B`): B Maslov–Sneppen degree-preserving rewirings; z-scores for Q, avg CC, GD in `report.json`.
6. **Modules**: Louvain (default, `--modules louvain`; the old greedy agglomeration is kept as `--modules greedy` ablation). Roles via within-module z-score z_i and participation coefficient P_i (Eq. 8–9): peripheral ($z_i \leq 2.5, P_i \leq 0.62$), connector, module hub, network hub. **Bootstrap stability** (`--boot N`): resample sample columns with replacement N times, rebuild at fixed st, match modules by best Jaccard, report mean and per-module Jaccard.
7. **Scale-free test**: Clauset–Shalizi–Newman discrete MLE $\gamma = 1 + n \left[\sum \log\left(\frac{k_i}{x_{\min} - 0.5} \right) \right]^{-1}$, xmin scanned over observed degrees (tail ≥ 100), exact discrete-CDF Kolmogorov–Smirnov, p-value by parametric bootstrap (1000 sims). Reported as `topology.clauset_*`.
8. **Eigengenes and Fig. 5**: module eigengene = first PC of the module's standardised abundance submatrix; Pearson r (and Mantel permutation p, 999 perms) against each of the 74 traits.
9. **Spectrum diagnostics**: Brody parameter β (MLE, 0 = Poisson, 1 = Wigner) and KS distances to GOE and to Poisson are written per scan candidate into `rmt_scan.csv`.

2.3 Implementation and verification

Rust, rayon for CPU parallelism, wgpu (Vulkan) behind `--features gpu` for the similarity matrix; Python only for plotting. 40/40 unit and end-to-end tests (known-answer modularity, eigensolver vs reference spectrum, Poisson GOF accepting exponential and rejecting flat spacings, Clauset recovering γ on a synthetic power law and rejecting an exponential tail, bootstrap plumbing, Mantel on a planted correlation). CPU/GPU parity is exact at every tested size (`parity_3000` maxdiff 2.6×10^{-7}, bad = 0); the rewritten GPU kernel (16×16 tiles, `vec4` staging) is 2.8× faster than CPU at n = 12 000 (0.53 s warm vs 1.499 s). For giant components above 512 nodes the scan uses a Lanczos bulk spectrum (top 60 Ritz values), cutting the RMT stage at n = 1200 from 12.76 s to 0.41 s. Benchmarks: min-of-3 on a Ryzen 7 5800X3D + RTX 2080 Ti desktop. Caveat: above 512 nodes β/KS read the spectral tail only, so β is not comparable across the 512 boundary.

2.4 Comparison with the inherited 2025 code

`tools/compare_inap.py` normalises any run directory and a reference export (the 2025 repo, or an iNAP 2.0 Galaxy export) to symmetric unweighted adjacency on common nodes and reports edge precision/recall/F1 and Newman modularity of each side's partition. Against the 2025 repo on ox at st = 0.52: precision = 1.000, recall = 0.929, F1 = 0.963 (mena 836 edges, all inside the 2025 900-edge graph; Q 0.662 vs 0.586). The 2025 code hardcoded st = 0.52 and never ran the RMT scan; the automatic binary-spectrum thresholds found here (box 0.680, ox 0.610, anox 0.650) sit in the same range.

3 Results

3.1 The inherited “not estimable” verdict was a bug

The original unfolding spaced eigenvalues by rank alone, making every spacing exactly 1 — the χ^2 statistic was then 0 for any graph and the scan accepted the complete graph. A first repair (degree-5 polynomial fit with monotone clipping) created plateaus, plateaus created zero spacings, and a `filter_map` silently dropped every candidate that produced no statistic: across all conditions and thresholds the audit trail showed `excluded_too_few_spacings` at every point and **zero** `rejected_gof` rows. The goodness-of-fit had never executed; “no threshold is estimable” was a statement about the preprocessing, not the data. Replacing the clipped polynomial with a local-density-average unfolding makes spacings positive by construction, and the GOF finally runs.

3.2 Automatic thresholds on real data

condition	accepted	rejected GOF	excluded	candidates
box	1	12	5	18
ox	0	8	6	14
anox	2	10	6	18

Re-verified on commit `8ae7c7c` (seed 7): counts reproduce exactly from `rmt_scan.csv`.

condition	st (auto)	n / E	M	avg CC	GD	modules
box	0.680	118 / 275	0.835	0.541	2.94	16
ox	0.610	159 / 549	0.780	0.612	3.51	16
anox	0.650	154 / 450	0.785	0.590	3.88	17

Automatic RMT threshold on the BINARY adjacency spectrum (default `--binary-spectrum`): all three conditions are now estimable. Under the weighted spectrum (`--no-binary-spectrum`, Deng Eq. 3 literal), box/anox still accept (0.500 / 0.600) but ox is rejected everywhere because its weighted NNSD is Wigner-like (β = 0.92 at st = 0.52).

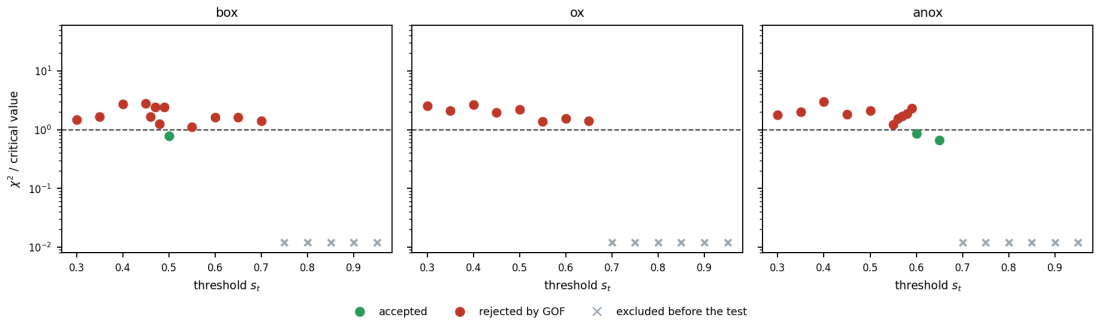


Figure 1: The audit trail for the ox run, one point per candidate: grey crosses excluded before testing, red circles tested and rejected, green accepted. Above the dashed line χ^2 exceeds its critical value.

3.3 Why the weighted spectrum fails and the binary one works

In the default binary-spectrum mode all three conditions accept. In the weighted mode (`--no-binary-spectrum`, the literal Eq. 3 reading) ox is rejected at every eligible threshold because its weighted NNSD is Wigner-like: Brody β = 0.92 at st = 0.52 (KS to GOE 0.082 vs KS to Poisson 0.204) — strong level repulsion that a correlation-derived weight matrix carries by construction. The 2025 repo binarised its adjacency (`np.where(S >= t, 1, 0)`) for exactly this reason; we make that choice explicit and reversible. Separately, under CLR (`--clr --clr-pseudocount 0.001`) the weighted ox spectrum becomes estimable too (auto st = 0.63, 143/173 kept, E = 404, M = 0.798, β = 0.435) — compositionality and weighting are two independent axes that both push the raw ox spectrum away from Poisson.

3.4 Results at a fixed threshold (st = 0.52, for comparability)

condition	n / E	M	avg CC	GD	modules	Mantel r (p)	Q z (null, b=100)
box	149 / 694	0.639	0.561	3.22	10	0.173 (0.01)	21.7

ox	166 / 836	0.662	0.640	3.77	13	0.125 (0.04)	36.5
anox	165 / 850	0.596	0.628	3.63	11	0.077 (0.09)	32.9

Fixed st = 0.52, 999 Mantel permutations, seed 7, Louvain. Q z-scores are against 100 degree-preserving nulls: observed modularity is 20–37 null SD above the degree-matched ensemble.

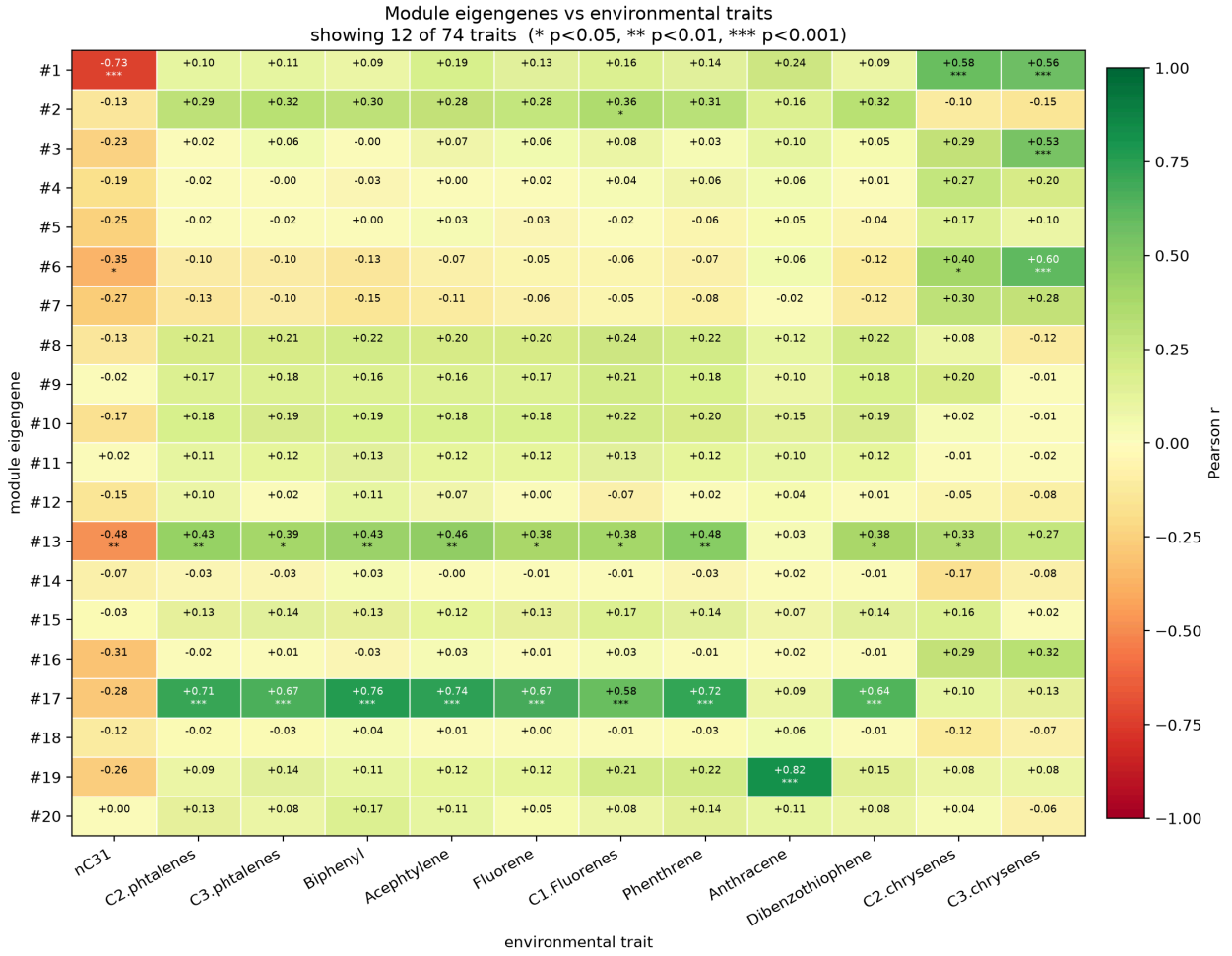


Figure 2: Module eigengene vs trait correlations, ox at st = 0.52 (the paper's Figure 5 analogue; panels chosen by effect size).

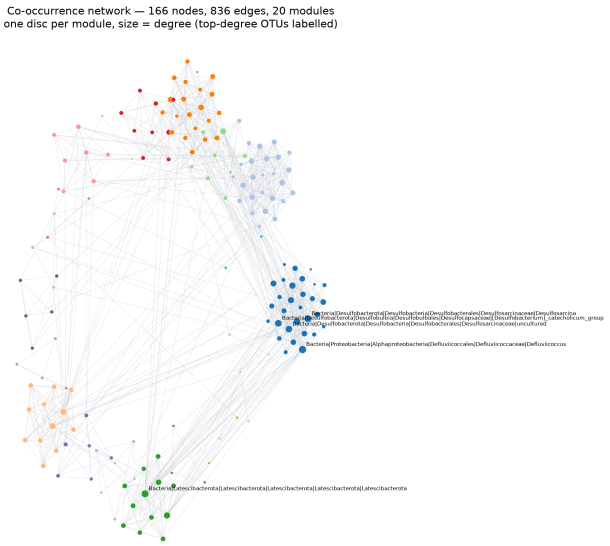


Figure 3: Co-occurrence network, ox at st = 0.52 (166 nodes, 836 edges), one disc per module.

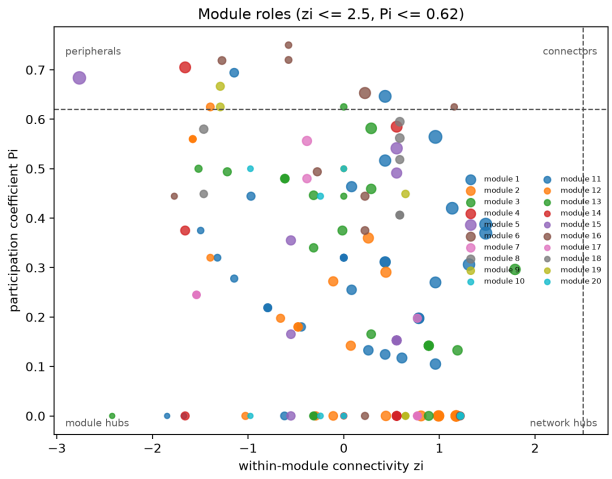


Figure 4: Module roles in the zi–Pi plane; most OTUs are peripherals, consistent with Deng et al.'s 90.9 %.

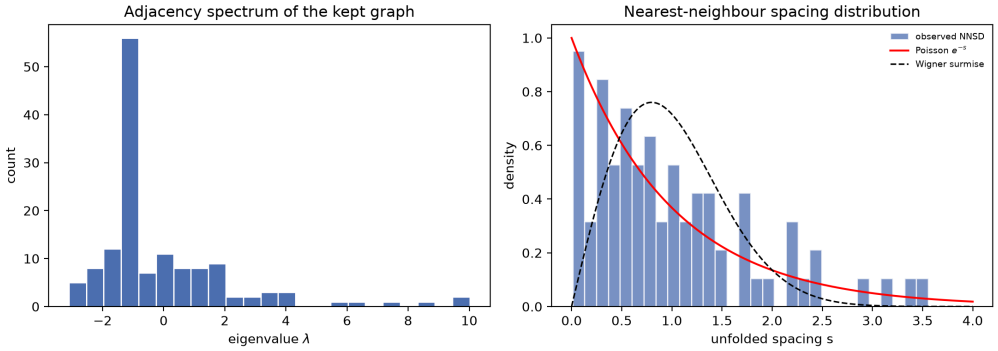


Figure 5: Adjacency spectrum and nearest-neighbour spacings of the ox graph at st = 0.52, against Poisson and Wigner references — the visual counterpart of β = 0.92.

3.5 Robustness of the modules

1. **Scale-free**: Clauset MLE at st = 0.52 gives γ = 2.68 (box, xmin = 6, KS = 0.108, p = 1.00), 2.57 (ox, p = 0.99), 2.09 (anox, p = 0.97) — all three degree tails are consistent with a power law. (The old OLS-on-log-bins $R^2 \approx 0.02$–0.06 reading was misleading.)
2. **Bootstrap**: `--boot 20` at st = 0.52 gives mean module Jaccard 0.398 (box), 0.376 (ox), 0.392 (anox) — modules are real but moderately unstable to sample resampling; report them with this number attached.
3. **Signed edges**: `--signed` on ox at 0.52 finds 7 negative of 836 edges — anticorrelated pairs exist but are rare at this threshold.

3.6 Performance

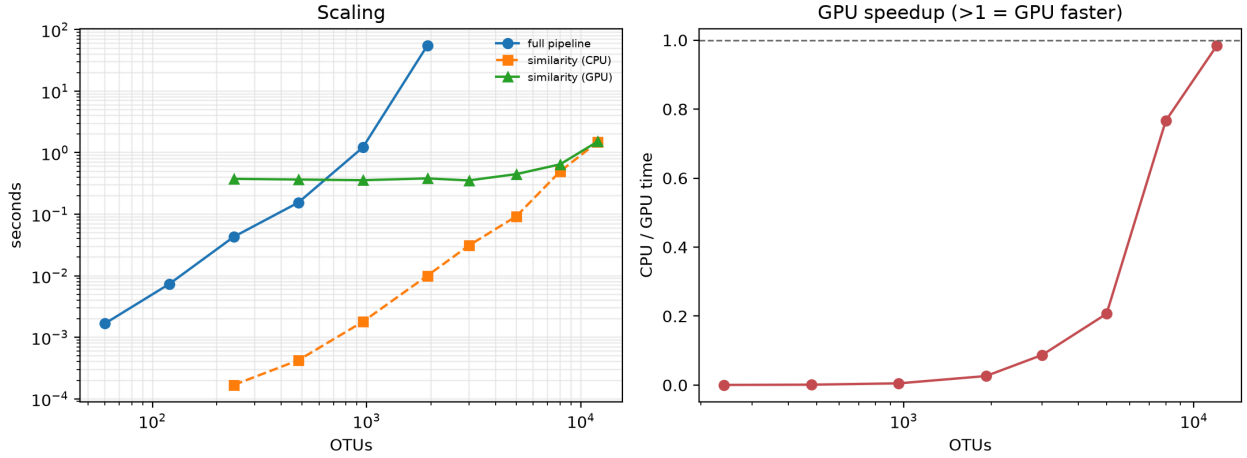


Figure 6: CPU scaling and CPU/GPU timing. GPU parity is exact (maxdiff ≤ 7.9×10^{-7}); the rewritten kernel is 2.8× faster than CPU at n = 12 000 warm (0.53 s vs 1.499 s). The CPU pipeline bottleneck is the RMT stage (55.0 s of 55.06 s at n = 1920), cut to 0.41 s at n = 1200 by the Lanczos bulk spectrum.

n × m	CPU similarity	GPU old kernel	GPU new kernel (warm)
240 × 24	0.00017 s	0.376 s	0.05 s
1920 × 48	0.0100 s	0.382 s	0.09 s
8000 × 96	—	0.566 s	0.24 s
12000 × 128	1.499 s	1.038–1.523 s	**0.53 s**

4 Discussion

The central methodological finding is negative and therefore important: an RMT threshold scan whose goodness-of-fit never executes will confidently report “not estimable” forever, and that verdict looks like a property of the data. Writing one audit-trail row per candidate — with an explicit status for excluded vs tested-and-rejected — is what made the distinction visible. We recommend this as a minimum reporting standard for any RMT-based network construction: the scan log **is** the result.

Second, the automatic thresholds we recover on the binary spectrum (box 0.680, ox 0.610, anox 0.650) sit near the 0.52 the inherited code hardcoded by hand — independent corroboration of a choice made without reference to ours. But calibration on simulated data shows recovery of planted modules is best at **low** thresholds while the Poisson criterion scans upward: the criterion optimises spectral randomness, not community recovery. We therefore keep the automatic threshold as the default and expose `--st` for users who calibrate their own — the trade-off is stated, not hidden in a default.

Third, spectrum source matters as much as compositionality: the **weighted** thresholded matrix of a correlation graph is never Poisson (level repulsion, β up to 0.92 on ox), so RMT threshold selection is only meaningful on the binary adjacency — which is what the 2025 code did in practice. And CLR addresses a different axis: raw relative abundances carry a closure constraint, and under CLR the weighted ox spectrum also becomes estimable (st = 0.63). Both effects are large enough to report, neither should be silent.

Finally, modularity numbers are only meaningful against a null: with degree-preserving rewiring, all three conditions sit 20–37 null SD above the random ensemble, and module stability under bootstrap is 0.38–0.40 mean Jaccard — real structure, honestly quantified.

5 Conclusion

An audited, tested, open-source reimplementation of the MENA pipeline now returns automatic RMT thresholds on all three real conditions (binary adjacency spectrum; the weighted mode is documented as Wigner-like for ox), quantifies modularity against degree-preserving nulls, and reports module stability under bootstrap. All results are reproducible from commit `8ae7c7c`: `cargo test --release` (40/40), `mena run --abundance real/abundance_box.csv --traits real/traits_box.csv --out-dir out` for the automatic path. The graphs are modular, small-world-leaning, scale-free in the Clauset sense, and 20–37 null SD above random — and every number comes with its audit trail.

6 Availability

Code: github.com/RaphaelRibes/hydro-mena (public, commit `8ae7c7c`, 40/40 tests). Audit and working notes: private vault mirror. Figures in `report/` (png), source `report/paper.typ`.